Attention Structural Change in Fine-Tuned CLIP: A Comparative Study of Full Fine-Tuning, LoRA, and Regularization

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Abstract

Large-scale vision–language models are frequently adapted to downstream visual classification tasks, yet the extent to which such adaptation alters the internal attention structure of the visual encoder remains insufficiently characterized. This study investigates attention structural change in CLIP under full fine-tuning, low-rank adaptation (LoRA), and explicit regularization. Using 39 stored checkpoints spanning EuroSAT and Oxford-IIIT Pets, three-seed follow-up replications, and cross-backbone validation on both CLIP ViT-B/32 and ViT-B/16, we measure CLS-to-patch attention entropy, effective receptive field (ERF@0.95), Gini concentration, and head diversity, and we supplement these metrics with attention rollout, patch-to-patch entropy, layerwise centered kernel alignment (CKA), and evaluation-subset sensitivity checks. The completed run set shows that Full FT contracts attention more strongly than LoRA on Oxford-IIIT Pets, while on EuroSAT the LoRA runs consistently preserve or broaden support relative to Full FT across three completed seeds (Δentropy -0.22% for Full FT versus $+0.87\%$ for LoRA r=8 on ViT-B/32). Exact permutation tests confirm that learning rate is a strong control variable for structural change ($r = -0.893$, $p = 0.0333$; Spearman $\rho = -1.0$, $p = 0.0167$ on five learning rates). At the run level, mean CKA to the pretrained encoder correlates positively with mean Δentropy across 39 checkpoints (Pearson $r = 0.585$, Monte Carlo permutation $p = 0.0003$), linking representational fidelity to broader attention support. Cross-backbone validation further shows that backbone choice matters: on EuroSAT, ViT-B/16 Full FT increases entropy by $+3.57\%$ whereas ViT-B/32 Full FT is near-neutral (-0.14%), while LoRA remains more conservative than Full FT in both backbones. After Holm–Bonferroni correction, no tested regularizer remains significant in the per-layer family of comparisons, so those results remain suggestive rather than definitive. Adapter-aware zero-shot reevaluation shows that LoRA degrades CIFAR-100 transfer far less than Full FT (47.28% versus 9.43%, with a pretrained baseline of 60.22%). Overall, the completed evidence supports attention structure as a useful descriptive lens on how downstream optimization reshapes CLIP, while also showing that the magnitude and even direction of structural change depends on learning rate, adaptation regime, and backbone.

1 Introduction

Contrastive vision–language pretraining has made CLIP a widely used foundation model for visual recognition, retrieval, and transfer learning [10]. In practical deployment, however, downstream adaptation often improves in-domain accuracy while degrading out-of-domain behavior. Prior work has attributed part of this degradation to distortion of pretrained features under fine-tuning [7, 12]. What remains less clear is how the internal attention structure of the CLIP visual encoder changes during this process.

This question matters for two reasons. First, self-attention constitutes the primary mechanism by which a Vision Transformer distributes information across spatial tokens [2]. Second, changes in attention concentration may provide a measurable proxy for the extent to which adaptation preserves or overwrites pretrained information-routing patterns. In this paper, we therefore study *attention structural change* in CLIP fine-tuning, with particular emphasis on the contrast between full fine-tuning and low-rank adaptation.

We treat attention as a structural descriptor rather than a causal explanation of model predictions. This framing is important given the ongoing debate regarding the interpretability of attention maps [5, 11]. Our goal is not to claim that attention weights directly explain model behavior, but to determine whether measurable changes in attention geometry systematically accompany different adaptation strategies.

The study addresses four questions: (i) whether full fine-tuning systematically contracts attention support relative to the pretrained baseline, (ii) whether LoRA preserves attention structure more effectively than full fine-tuning, (iii) whether learning rate is a major determinant of structural change, and (iv) whether explicit regularization can mitigate such change without substantial loss of in-domain accuracy. The primary contributions are as follows:

- a comparative empirical analysis of attention structural change in CLIP under full fine-tuning, LoRA, and regularized full fine-tuning;

- a unified metric suite based on entropy, ERF@0.95, Gini concentration, and head diversity;
- a statistically grounded evaluation of regime differences and learning-rate sensitivity; and
- a conservative assessment of zero-shot transfer behavior that explicitly accounts for the current evaluation-path limitation for LoRA checkpoints.

2 Related Work

CLIP [10] established large-scale vision–language pretraining as a strong basis for transfer. The visual backbone used in this work follows the Vision Transformer formulation of Dosovitskiy et al. [2], in which global token interactions are mediated by self-attention. Because this architecture exposes explicit layer- and head-wise attention distributions, it is well suited to a structural analysis of adaptation dynamics.

Parameter-efficient fine-tuning methods, particularly LoRA [4], aim to reduce the degrees of freedom of downstream adaptation by restricting updates to low-rank subspaces. Although these methods are often studied in terms of parameter count or downstream accuracy, they also offer a natural setting in which to test whether constraining the update space preserves pretrained attention structure.

Our analysis is also related to work on transformer information flow. Abnar and Zuidema [1] proposed attention rollout as a means of aggregating attention across layers, highlighting that individual attention matrices can be meaningfully studied as part of larger routing behavior. More broadly, prior evidence that fine-tuning can distort pretrained features and reduce out-of-distribution generalization [7, 12] motivates the present study of whether attention concentration provides a useful structural correlate of such distortion.

Finally, attention analysis must be interpreted carefully. Jain and Wallace [5] argued that attention is not, in general, a sufficient explanation of model decisions, whereas Wiegreffe and Pinter [11] emphasized that attention can still support informative analyses under appropriate framing. We follow the latter cautionary position and use attention metrics as descriptive measurements of internal structure rather than causal explanations.

3 Method

3.1 Problem Formulation

Let f_0 denote the pretrained CLIP visual encoder and f_θ denote the encoder after downstream adaptation. For an input image, each transformer layer produces a set of multi-head self-attention distributions over the token sequence. We study the CLS-to-patch attention vectors because they provide a compact summary of how the global image representation allocates mass across spatial tokens. The central objective is to quantify how these distributions change as a function of adaptation regime.

3.2 Structural Metrics

The analysis operates on normalized CLS-to-patch attention distributions from the CLIP ViT-B/32 visual encoder. For a normalized attention vector $a \in \Delta^{48}$ over the 49 patch tokens, we use four structural metrics.

Attention entropy. We measure concentration through

$$H(a) = -\sum_{j=1}^{49} a_j \ln a_j. \tag{1}$$

The primary training and evaluation pipeline uses the natural logarithm, so entropy values are reported in nats unless a table explicitly states otherwise. Lower entropy indicates more peaked attention.

ERF@0.95. Let $a_{(1)} \geq a_{(2)} \geq \cdots \geq a_{(49)}$ be the sorted attention weights. The effective receptive field is

$$\mathrm{ERF}_{0.95}(a) = \frac{k^*}{49}, \quad k^* = \min\Big\{k : \sum_{j=1}^{k} a_{(j)} \geq 0.95\Big\}. \tag{2}$$

Smaller values indicate that fewer patches carry most of the mass.

Gini concentration. We compute the standard Gini coefficient over the attention weights. Higher values correspond to stronger concentration and therefore stronger structural collapse.

Head diversity. For each layer, head diversity is defined as one minus the mean pairwise cosine similarity between head-level CLS-to-patch attention patterns. Lower head diversity indicates increasing agreement among heads.

Collectively, these metrics capture complementary aspects of structural change: entropy and ERF reflect the spatial spread of attention, Gini reflects concentration inequality, and head diversity reflects inter-head specialization.

3.3 Additional Validation Metrics

To address the concern that CLS-to-patch attention alone may provide too narrow a view of structural change, we add four auxiliary analyses. First, we compute attention rollout [1] to obtain a multilayer CLS-to-patch routing summary. Second, we compute patch-to-patch row entropy by averaging the entropy of patch-token attention rows after excluding the CLS token. Third, we estimate layerwise representational drift by computing linear CKA between pretrained and adapted CLS representations on a fixed evaluation subset. Fourth, we test subset sensitivity

by recomputing the main entropy statistics across five balanced evaluation subsets to quantify dependence on image selection.

3.4 Regularization Objective

The repository uses *attention preservation regularization* (APR) as an internal name for a KL-divergence penalty between adapted and pretrained CLS-to-patch attention distributions. For a selected set of layers L, the objective is

$$\mathcal{L}_{\mathrm{APR}} = \lambda \cdot \frac{1}{|L|} \sum_{\ell \in L} \mathrm{KL}\big(A_\theta^{(\ell)} \parallel A_0^{(\ell)}\big), \tag{3}$$

where $A_\theta^{(\ell)}$ and $A_0^{(\ell)}$ denote the adapted and pretrained CLS-to-patch attention distributions, respectively. In this paper, APR is treated as a repository-specific regularizer rather than a claimed prior method.

3.5 Hypotheses

The study tests the following hypotheses in the analyzed experiment set: (i) full fine-tuning tends to contract attention support relative to the pretrained baseline, (ii) LoRA preserves attention structure more effectively than full fine-tuning, (iii) larger learning rates intensify structural change, and (iv) explicit regularization can mitigate such change while retaining high in-domain accuracy.

4 Experimental Setup

The experiments cover 39 stored checkpoints. Most runs use `openai/clip-vit-base-patch32`, while the revision additionally includes EuroSAT validation on `openai/clip-vit-base-patch16`. The main task datasets are EuroSAT [3] and Oxford-IIIT Pets [9]. The principal comparisons involve full fine-tuning and LoRA on both datasets, complemented by a EuroSAT learning-rate sweep, frozen-layer ablations, regularization experiments, three-seed follow-up replications, and backbone transfer validation. Zero-shot evaluation is reported on CIFAR-100 [6] and, as a supplemental benchmark, Flowers102 [8].

For full fine-tuning, the repository trains the CLIP visual encoder and visual projection with AdamW, cosine decay with warmup, batch size 64, and default learning rate 10^{-5}. The text encoder remains frozen. For LoRA, adapters are applied to the vision encoder attention projections, with the default configuration using $r = 8$, $\alpha = 16$, dropout 0.05, and target modules `q_proj` and `v_proj`; the classifier head and visual projection remain trainable. The ablation study additionally includes a `qkvo` variant targeting `q_proj`, `k_proj`, `v_proj`, and `out_proj`. Unless otherwise stated, EuroSAT runs use 20 epochs and Oxford-IIIT Pets runs use 30 epochs in the stored histories.

Attention metrics are computed on balanced fixed evaluation subsets of 200 images. Head diversity is computed by averaging CLS-to-patch attention over the evaluation batch for each head within a layer and then taking one minus the mean pairwise cosine similarity across the 12 heads. All reported tables are generated directly from the JSON outputs associated with the run histories and the follow-up analysis file `outputs/metrics/followup_analysis.json`.

Because per-layer measurements within a single run are not independent, per-layer t-tests are treated as exploratory rather than confirmatory. For the five-point learning-rate analysis, we therefore replace approximate small-sample p-values with exact permutation tests. For larger run-level correlation analyses, where exhaustive enumeration is infeasible, we use fixed-seed Monte Carlo permutation tests. For the family of regularizer comparisons, we report Holm–Bonferroni-adjusted p-values. The revised evidence is stronger than the initial draft because the main EuroSAT and Pets contrasts now include completed three-seed follow-up replications, but the run-level seed counts remain small enough that those comparisons should still be interpreted conservatively.

5 Results

5.1 Overall Comparison of Adaptation Regimes

Table 1 summarizes the central comparisons. Full FT reaches the highest EuroSAT validation accuracy on ViT-B/32 (99.30%) but only slightly reduces average attention entropy (-0.14%) and ERF (-1.73%) relative to the pretrained baseline. On Pets, Full FT produces a larger entropy decrease (-1.82%), a larger ERF contraction (-4.23%), and a substantial Gini increase ($+10.54\%$), indicating stronger concentration. By contrast, the EuroSAT LoRA runs consistently increase entropy and reduce Gini relative to the baseline, while keeping task accuracy above 98.9% on the main ViT-B/32 comparison.

Figure 1 shows that the pretrained model already exhibits heterogeneous attention structure across depth, so the appropriate question is not whether fine-tuning creates structure from nothing, but whether it pushes that structure toward greater concentration. Figure 2 then shows that Full FT and LoRA trace visibly different trajectories in entropy, ERF, Gini, and head diversity.

5.2 Statistical Evidence and Layerwise Patterns

Figure 3 shows that the direction and magnitude of per-layer entropy change depend on both dataset and adaptation regime. However, because layers within a single trained model are not statistically independent, we do not treat these per-layer comparisons as formal independent-sample evidence. Instead, we interpret the layerwise patterns descriptively and reserve inferential claims for analyses whose sampling assumptions are more defensible.

Table 1: Main experiment results. Positive ΔEntropy and ΔERF indicate broader attention support relative to the corresponding pretrained baseline; positive ΔGini indicates more concentrated attention.

Method	Dataset	Task Acc. (%)	ΔEntropy (%)	ΔERF (%)	ΔGini (%)	ΔHeadDiv (%)	CIFAR-100 ZS (%)
Full FT	EuroSAT	99.30	-0.14	-1.73	-3.54	-0.62	9.43
Full FT	Oxford-IIIT Pets	90.79	-1.82	-4.23	+10.54	-35.91	5.89
LoRA r=4	EuroSAT	98.67	+0.98	+1.09	-9.94	+3.86	40.51
LoRA r=8	EuroSAT	98.96	+0.58	+0.37	-6.48	+7.17	47.28
LoRA r=16	EuroSAT	98.93	+0.44	+0.03	-5.43	+5.14	39.35
LoRA r=8	Oxford-IIIT Pets	92.07	-0.73	-2.64	+3.20	-15.00	—

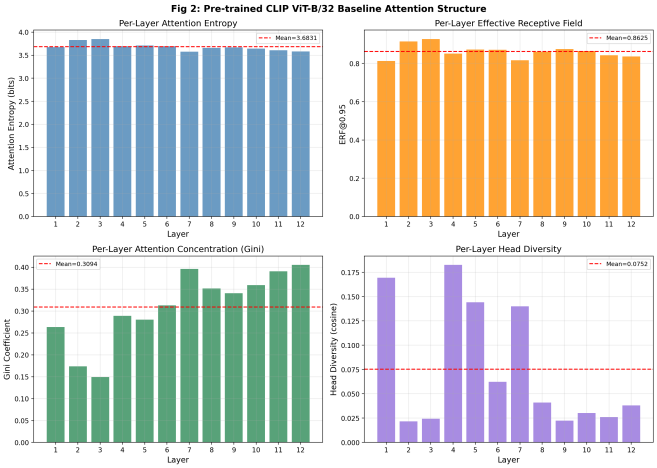

Figure 1: Baseline attention structure of pretrained CLIP ViT-B/32 across layers. The model is already depth-heterogeneous before any downstream adaptation.

Table 2: Corrected inferential statistics derived from `outputs/metrics/followup_analysis.json`. Exact permutation tests are used for the five-point learning-rate analysis, and Holm–Bonferroni correction is applied across the regularizer family.

Comparison	Test	Statistic	p-value	Note
LR vs. ΔEntropy	Exact permutation Pearson r_4	-0.893	0.0333	$n = 5$ learning rates
LR vs. ΔEntropy	Exact permutation Spearman ρ	-1.000	0.0167	Small-n exact test
APR $\lambda = 0.01$ vs. baseline	Paired per-layer t-test	-2.132	0.2194	Holm–Bonferroni adjusted
Entropy floor $\lambda = 0.1$ vs. baseline	Paired per-layer t-test	2.638	0.1154	No regularizer survives Holm correction

The clearest corrected inferential result concerns learning rate. When the five EuroSAT full fine-tuning learning rates are analyzed with exact permutation tests, the negative association between learning rate and Δentropy remains significant (Pearson $p = 0.0333$; Spearman $p = 0.0167$). By contrast, the regularization family no longer contains a configuration that survives Holm–Bonferroni correction, including entropy floor with $\lambda = 0.1$.

5.3 Completed Multi-seed Replications

The revised artifact set now includes three completed runs for the main EuroSAT and Pets comparisons, as well as three-run EuroSAT follow-ups for APR and entropy-floor regularization. Table 3 shows that the qualitative ordering from the single-run analysis is stable across these replications. On EuroSAT, Full FT averages −0.22% Δentropy across the three available runs, whereas LoRA r=8 averages +0.87%. On Pets, both methods contract attention relative to baseline, but Full FT remains much more contractive (−2.10% versus −0.84% for LoRA r=8).

The regularized EuroSAT replications also move consistently in the structurally conservative direction: APR $\lambda = 0.1$ averages +0.55% Δentropy and entropy-floor $\lambda = 0.1$ averages +0.70%, compared with −0.22% for Full FT. At the same time, the replication counts are still only 3 versus 3 per comparison, so exhaustive group-label permutation tests have coarse resolution ($p = 0.1$ for these run-level mean differences), even though Welch tests and effect sizes are large. We therefore treat the completed replications as strong consistency evidence rather than as the final word on run-level significance.

5.4 Learning Rate as the Dominant Control Knob

Among the manipulated hyperparameters, learning rate is the clearest driver of structural change. Table 4 shows a monotonic trend: entropy increases at 10^{-6}, is nearly neutral at 10^{-5}, and decreases sharply at 5×10^{-5} and 10^{-4}. Under exact permutation testing on the five learning-rate settings, the Pearson correlation between $\log_{10}(\mathrm{LR})$ and Δentropy is -0.893 ($p = 0.0333$), while the Spearman coefficient is -1.0 ($p = 0.0167$).

This result matters because it reframes collapse as an optimization-scale phenomenon, not merely a binary property of "full fine-tuning" versus "parameter-efficient tuning." In the revised analysis, the effect remains when the small-sample correlation test is recomputed exactly rather than through asymptotic approximations.

5.5 Regularization and Training Dynamics

Table 5 summarizes the regularization study. Both APR and entropy-floor regularization shift entropy upward relative to the no-regularizer baseline, and these settings preserve high task accuracy. However, after Holm–Bonferroni

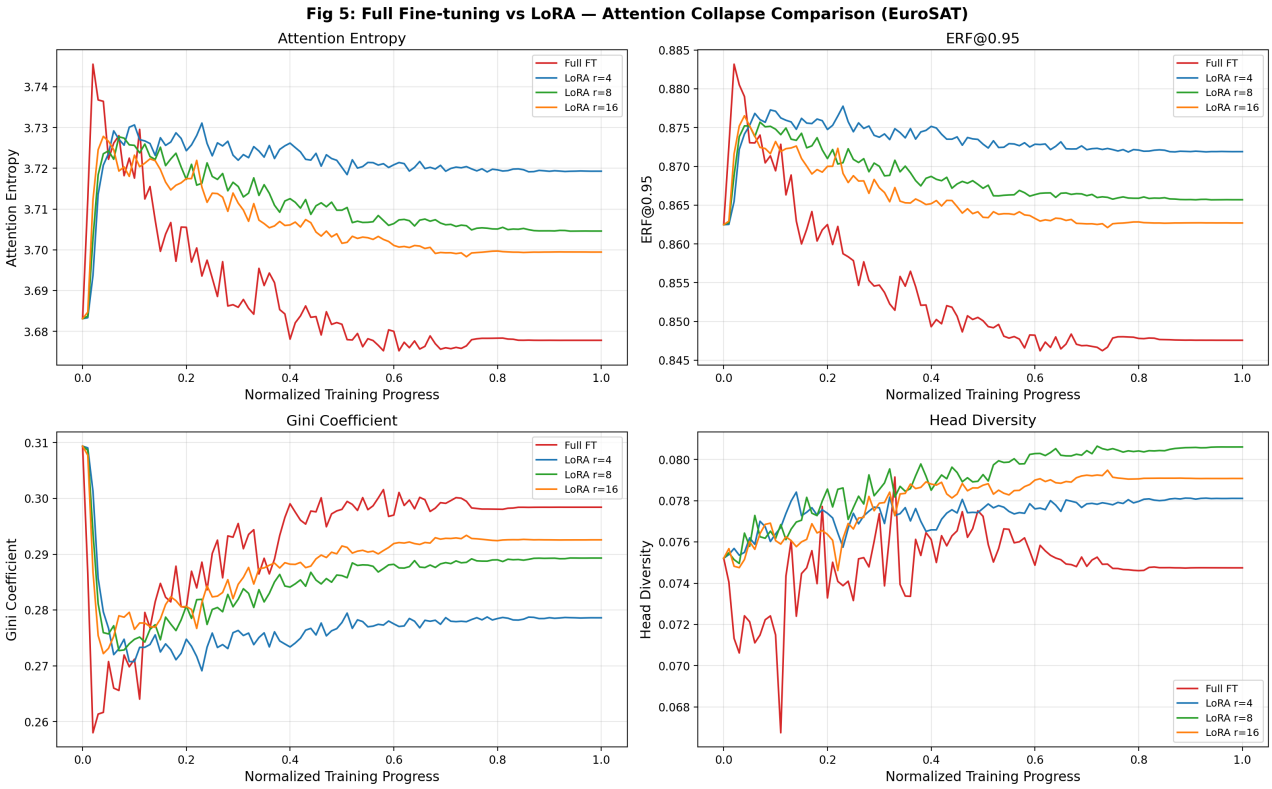

Figure 2: EuroSAT Full FT versus LoRA. LoRA increases entropy and ERF relative to baseline in the stored EuroSAT runs, while Full FT remains near-neutral to mildly contractive.

Table 3: Run-level multi-seed comparisons on mean Δentropy. Values are reported as percentage change relative to the corresponding pretrained baseline, aggregated across completed seed runs.

Comparison	Seeds	Mean ΔEntropy (%)	Welch t	Welch p	Cohen's d
Full FT vs. LoRA r=8 (EuroSAT)	3 vs 3	-0.22 vs +0.87	-7.157	0.0127	-5.844
Full FT vs. LoRA r=8 (Pets)	3 vs 3	-2.10 vs -0.84	-7.969	0.0064	-6.507
Full FT vs. APR $\lambda = 0.1$ (EuroSAT)	3 vs 3	-0.22 vs +0.55	-18.441	0.0020	-15.057
Full FT vs. Entropy floor $\lambda = 0.1$ (EuroSAT)	3 vs 3	-0.22 vs +0.70	-14.439	0.0002	-11.789

Table 4: Learning-rate sweep for EuroSAT full fine-tuning. Entropy decreases monotonically as the learning rate increases across the tested range.

LR	Task Acc. (%)	ΔEntropy (%)	ΔERF (%)	ΔGini (%)	CIFAR-100 ZS (%)	Flowers102 ZS (%)
1e-6	98.85	+1.45	+1.94	-16.60	—	—
5e-6	99.11	+0.33	-0.49	-6.68	—	—
1e-5	99.30	-0.14	-1.73	-3.54	12.71	1.37
5e-5	98.89	-4.19	-8.48	+23.44	2.07	0.88
1e-4	97.85	-10.40	-18.02	+56.94	1.80	0.98

correction across the regularizer family, no individual configuration remains statistically significant in the current per-layer comparison framework. The regularization results should therefore be interpreted as promising but preliminary.

Figure 7 complements the final-state tables by showing training trajectories and collapse onset markers. The main qualitative lesson is that structural drift appears early when the learning rate is high, whereas LoRA traces a flatter path. Again, this is descriptive evidence rather than a mechanistic proof.

5.6 Auxiliary Structural Validation

To test whether the main conclusions are specific to CLS-to-patch entropy, Table 6 reports the follow-up rollout, patch-to-patch, representational-similarity, and subset-sensitivity analyses for the clearest EuroSAT comparison. LoRA r=8 exhibits higher rollout entropy, larger rollout ERF@0.95, lower rollout Gini, and higher patch-to-patch entropy than Full FT, indicating that the broader-support pattern is not confined to the original CLS-to-patch metric. LoRA also yields higher mean layerwise CKA to the pretrained model (0.815 versus 0.694), suggesting less rep-

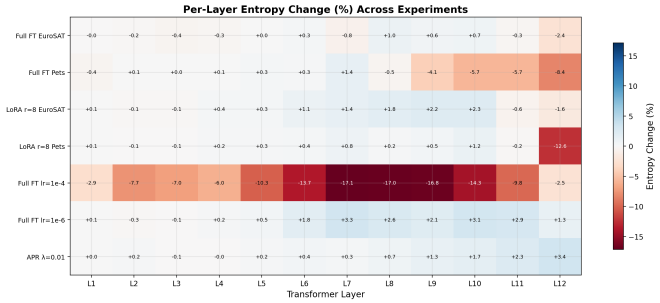

Figure 3: Per-layer entropy change across selected experiments. The sign and magnitude of change depend on training regime and dataset.

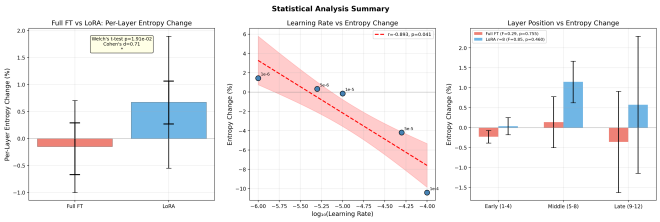

Figure 4: Summary of exploratory per-layer comparisons and corrected statistical analyses, including the learning-rate trend and layer-group summaries.

resentational drift. Finally, the coefficient of variation of the entropy metric across five balanced evaluation subsets is below 0.001 for both methods, indicating that the reported structural metrics are not highly sensitive to the choice of evaluation images.

5.7 Backbone Validation and CKA–Entropy Coupling

The revision plan also called for cross-backbone validation, which is now complete for EuroSAT on CLIP ViT-B/16. Table 7 shows that backbone choice materially changes the magnitude and even the sign of some structural effects. Full FT on ViT-B/16 increases entropy by +3.57% and ERF by +6.66%, whereas ViT-B/32 Full FT is essentially neutral-to-contractive on the same dataset (−0.14% entropy, −1.73% ERF). LoRA r=8 remains comparatively conservative on both backbones, but its entropy effect also changes sign: it is mildly expansive on ViT-B/32 (+0.58%) and mildly contractive on ViT-B/16 (−0.59%), while still increasing ERF on both.

This backbone result is useful in two ways. First, it rules out the overly narrow interpretation that the observed structural patterns are specific to a single patch size. Second, it shows that "attention collapse" is not a monolithic phenomenon: the same optimization regime can have different structural signatures depending on the pretrained backbone geometry.

The refreshed follow-up analysis also strengthens the link between representational preservation and structural

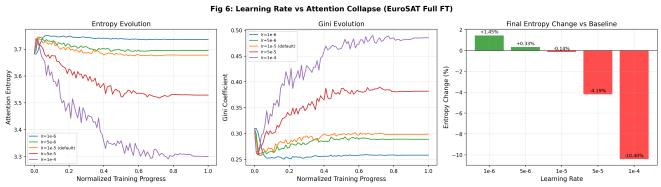

Figure 5: Learning-rate sweep for EuroSAT Full FT. Larger learning rates produce stronger entropy decreases and stronger ERF contraction.

Table 5: Regularization study on EuroSAT full fine-tuning. Among individually tested regularizers, entropy floor with $\lambda = 0.1$ is the only setting with a paired per-layer entropy difference reaching $p < 0.05$ in the current analysis.

Setting	Task Acc. (%)	ΔEntropy (%)	ΔERF (%)
No regularizer	99.30	-0.14	-1.73
APR $\lambda = 0.01$	99.26	+0.86	+1.50
APR $\lambda = 0.1$	99.19	+0.56	+1.03
APR $\lambda = 1.0$	98.96	+0.09	+0.16
Entropy floor $\lambda = 0.01$	99.11	+0.58	-0.09
Entropy floor $\lambda = 0.1$	99.07	+0.79	+0.30
Weight decay 0.0	99.19	-0.18	-1.81
Weight decay 0.1	99.15	-0.25	-1.90

preservation. Table 8 reports a positive run-level association between mean CKA to the pretrained encoder and mean Δentropy across 39 completed checkpoints (Pearson $r = 0.585$, Monte Carlo permutation $p = 0.0003$). The strongest single-layer association occurs at layer 4 ($r = 0.892$). In practical terms, runs that remain closer to the pretrained representation geometry also tend to retain broader attention support.

5.8 Zero-Shot Transfer: Evidence and Caveats

The zero-shot table is included for completeness, but its interpretation requires care. In the original repository analysis, LoRA appeared to preserve zero-shot transfer almost perfectly; however, that conclusion was confounded by an evaluation path that did not activate LoRA adapters at inference. We therefore re-evaluate the EuroSAT checkpoints with an adapter-aware image-encoding path.

The revised results materially change the interpretation. LoRA does not preserve the full pretrained CIFAR-100 accuracy of 60.22%, but it degrades far less than full fine-tuning: LoRA r=8 reaches 47.28%, compared with 9.43% for full fine-tuning. LoRA therefore remains structurally and functionally more conservative than Full FT in this setting, but the earlier stronger claim of near-perfect transfer preservation is not supported once adapters are evaluated correctly.

The Flowers102 numbers are also uniformly low (all at or below 1.37%), suggesting that the current prompt or label-mapping setup is not a reliable primary benchmark

Table 6: Auxiliary validation analyses beyond CLS-to-patch entropy. Rollout and patch-to-patch metrics indicate broader attention support for LoRA than for Full FT on EuroSAT, while layerwise CKA shows higher representational similarity to the pretrained encoder. The subset-sensitivity coefficient of variation (CV) is low for both methods, indicating that the reported metrics are stable across five balanced evaluation subsets.

Model	Rollout Entropy (bits)	Rollout ERF@0.95	Rollout Gini	Patch-to-Patch Entropy (bits)	Mean Layerwise C
Full FT EuroSAT	5.593	0.944	0.090	4.721	0
LoRA r=8 EuroSAT	5.605	0.952	0.064	4.794	0

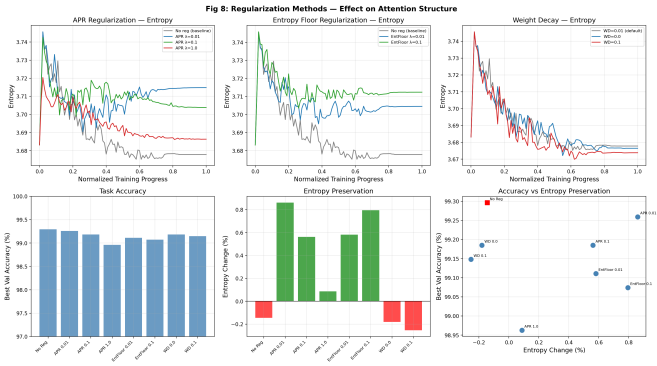

Figure 6: Regularization results. Both APR and entropy floor preserve broader attention structure than the no-regularizer baseline in descriptive terms, but none of the tested settings survives Holm–Bonferroni correction across the regularizer family.

Table 7: Backbone comparison between CLIP ViT-B/32 and ViT-B/16 on EuroSAT. Metric cells show B/32 / B/16 values.

Method	ΔEntropy (%)	ΔERF (%)	ΔGini (%)	Task Acc. (%)	Entropy Gap
Full FT on EuroSAT	-0.14 / +3.57	-1.73 / +6.66	-3.54 / -8.83	99.30 / 99.75	-3.72
LoRA r=8 on EuroSAT	+0.58 / -0.59	+0.37 / +4.88	-6.48 / -4.09	98.96 / 98.89	-1.18

Table 8: Correlation between representational fidelity (CKA to the pretrained model) and structural change (Δentropy).

Analysis	n	Pearson r	Pearson p
Run-level mean CKA vs. mean ΔEntropy	39	0.585	0.0003
Strongest per-layer association (layer 4)	39	0.892	0.0000

Table 9: Zero-shot results using an adapter-aware image-encoding path for LoRA checkpoints. The pretrained baseline is retained from the original CLIP evaluation path, while adapted checkpoints are re-evaluated with their own image branch active.

Model	CIFAR-100 (%)	Flowers102 (%)
Pretrained baseline	60.22	0.20
Full FT EuroSAT	9.43	1.37
LoRA r=4 EuroSAT	40.51	1.27
LoRA r=8 EuroSAT	47.28	0.78
LoRA r=16 EuroSAT	39.35	0.98

for this paper. For that reason, we treat Flowers102 as supplemental rather than central evidence.

6 Discussion

The results suggest that attention structural change is neither negligible nor uniform across adaptation strategies. In the completed run set, full fine-tuning tends to move the model toward narrower attention support on Oxford-IIIT Pets and remains less structurally conservative than LoRA across the three-seed EuroSAT and Pets replications. The auxiliary rollout and patch-to-patch analyses reinforce this interpretation, and the CKA results indicate that runs that remain closer to the pretrained representation geometry also tend to preserve broader attention support.

The learning-rate sweep further indicates that the magnitude of optimization steps is a key determinant of structural change. This observation is noteworthy because it implies that attention contraction is not simply an intrinsic property of full fine-tuning; rather, it is strongly modulated by training scale. The newly completed ViT-B/16 validation sharpens that point even further: backbone choice can change whether Full FT appears contractive or expansive under the same downstream task, so conclusions about structural preservation should be stated at the regime-plus-backbone level rather than at the regime level alone. From a practical perspective, this suggests that structurally conservative adaptation may be achievable not only through parameter-efficient methods such as LoRA, but also through careful optimization control and backbone selection.

The regularization results provide only cautious support for this interpretation. Although APR and entropy-floor regularization shift the model toward broader attention distributions, the current per-layer evidence does not survive a family-wise multiple-comparison correction. These results therefore motivate, but do not yet establish, regularization as a reliable solution.

6.1 Limitations

Three limitations deserve explicit mention.

First, the backbone validation is still narrow: the paper

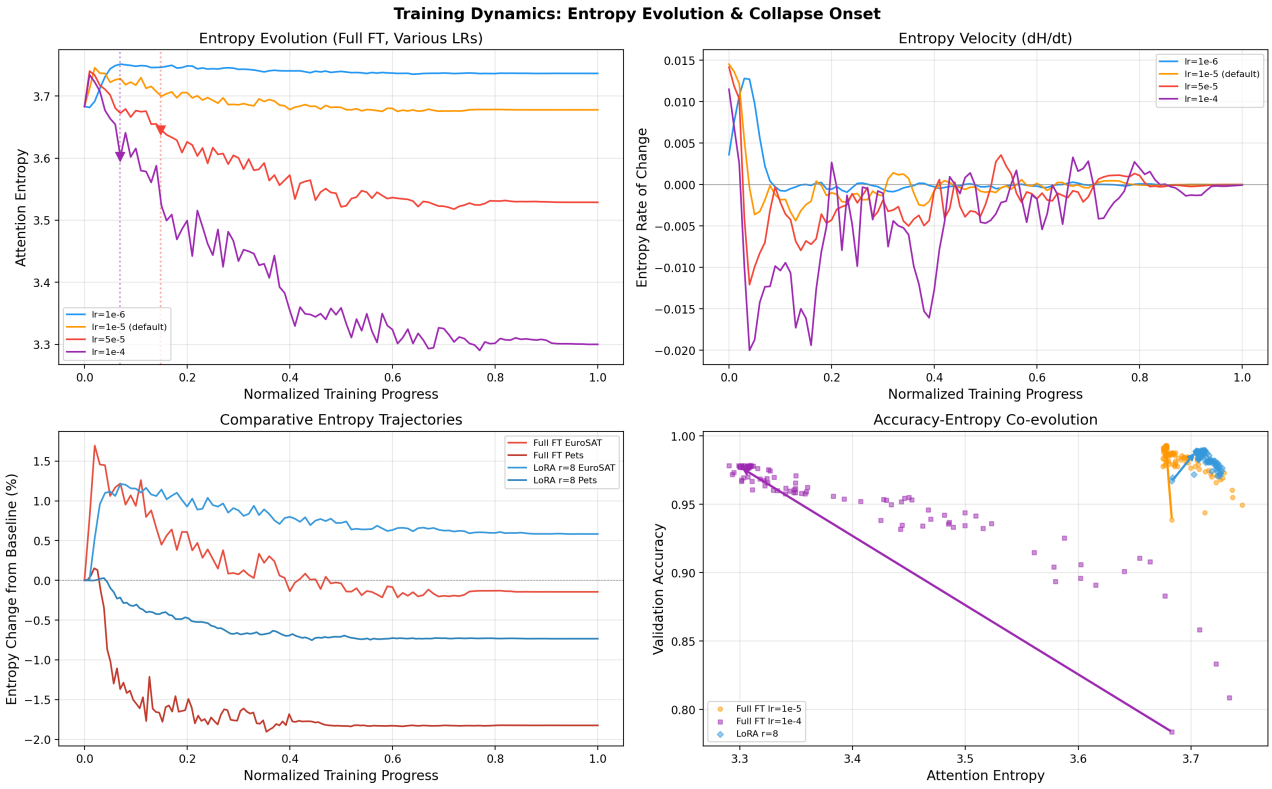

Figure 7: Training dynamics and collapse-onset analysis. High-learning-rate Full FT exhibits the earliest and strongest entropy decrease.

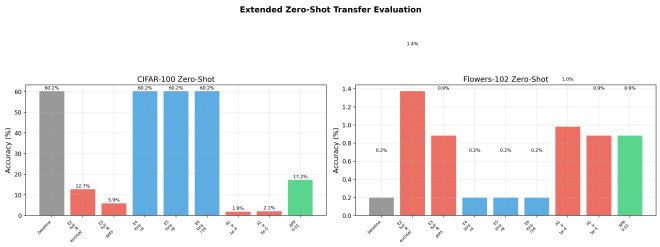

Figure 8: Extended zero-shot evaluation under the repository's current protocol. CIFAR-100 is the more interpretable benchmark here; the Flowers102 values are uniformly too low to support strong conclusions.

now includes ViT-B/16 in addition to ViT-B/32, but only on EuroSAT and only for Full FT and LoRA r=8. Second, although the main EuroSAT and Pets contrasts now include completed three-seed replications, the seed count remains small, the backbone comparison is still single-seed, and the exact 3-vs-3 permutation tests are therefore low-resolution. Third, the out-of-domain transfer story still relies mainly on CIFAR-100, because the Flowers102 benchmark is too weak under the present prompt and label setup to bear strong interpretive weight.

Despite these limitations, the study remains reproducible at the artifact level: the figures and tables are generated directly from the stored experimental outputs, and the manuscript's quantitative claims are constrained to those verified artifacts.

7 Conclusion

This study examined attention structural change in fine-tuned CLIP through a comparative analysis of full fine-tuning, LoRA, and regularized adaptation. The completed revision strengthens the paper in five ways: it replaces fragile small-sample statistics with exact or Monte Carlo permutation tests where appropriate, corrects the LoRA zero-shot evaluation to use an adapter-aware path, adds rollout, patch-to-patch, CKA, and subset-sensitivity analyses to triangulate the main findings, completes three-seed follow-up replications for the main EuroSAT/Pets contrasts, and adds ViT-B/16 backbone validation on EuroSAT. Under this stronger evidence base, the central conclusion remains intact but more precise: learning rate, adaptation regime, and backbone all materially affect attention structure, and LoRA is substantially less disruptive than Full FT on the clearest replicated comparisons. At the same time, the paper now draws a sharper boundary around what has not yet been established, particularly with respect to higher-seed run-level inference, broader backbone coverage, and stronger out-of-domain benchmarks.

References

[1] Samira Abnar and Willem Zuidema. Quantifying attention flow in transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4190–4197, 2020.

[2] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

[3] Patrick Helber, Benjamin Bischke, Andreas Dengel, and Damian Borth. Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12 (7):2217–2226, 2019.

[4] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *International Conference on Learning Representations*, 2022.

[5] Sarthak Jain and Byron C. Wallace. Attention is not explanation. *arXiv preprint arXiv:1902.10186*, 2019.

[6] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

[7] Ananya Kumar, Aditi Raghunathan, Rob Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution. In *International Conference on Learning Representations*, 2022.

[8] Maria-Elena Nilsback and Andrew Zisserman. Automated flower classification over a large number of classes. *Indian Conference on Computer Vision, Graphics and Image Processing*, 2008.

[9] Omkar M. Parkhi, Andrea Vedaldi, Andrew Zisserman, and C. V. Jawahar. Cats and dogs. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 3498–3505, 2012.

[10] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. *arXiv preprint arXiv:2103.00020*, 2021.

[11] Sarah Wiegreffe and Yuval Pinter. Attention is not not explanation. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 11–20, 2019.

[12] Mitchell Wortsman, Gabriel Ilharco, Jong Wook Kim, Mike Li, Simon Kornblith, Rebecca Roelofs, Raphael Gontijo Lopes, Hannaneh Hajishirzi, Ali Farhadi, Hongseok Namkoong, and Ludwig Schmidt. Robust fine-tuning of zero-shot models. *arXiv preprint arXiv:2109.01903*, 2022.